

Table 4: Search Model Estimates on Real Data

	NNE		SMLE ($\lambda = 3$)		SMLE ($\lambda = 7$)		SMLE ($\lambda = 10$)	
Simul. burden ($\times 10^4$)	1.00		9.09		8.28		8.24	
Compute time (mins)	3.83		33.27		30.34		30.25	
Stars, β_1	0.215	(.078)	0.218	(.109)	0.111	(.063)	0.081	(.047)
Review, β_2	-0.089	(.107)	-0.046	(.188)	-0.005	(.106)	0.001	(.076)
Location, β_3	0.111	(.108)	0.178	(.196)	0.080	(.108)	0.056	(.077)
Chain, β_4	0.018	(.093)	-0.012	(.119)	-0.007	(.067)	-0.007	(.049)
Promotion, β_5	0.183	(.091)	0.105	(.108)	0.050	(.062)	0.037	(.045)
Price, β_6	-0.240	(.080)	-0.390	(.117)	-0.208	(.071)	-0.155	(.052)
Outside, η	4.038	(.594)	3.670	(.927)	2.803	(.510)	2.557	(.361)
Search, δ_0	-2.953	(.259)	-0.307	(.260)	-1.290	(.191)	-1.567	(.151)
Search, δ_1	0.049	(.036)	-0.147	(.093)	-0.083	(.067)	-0.061	(.053)

Notes: Numbers in parentheses are asymptotic standard errors (for SMLE) and estimates of statistical accuracy outputted by neural net (for NNE). All reported numbers are averaged across 100 estimation runs on the real data (estimates vary slightly between estimation runs because SMLE and NNE are both simulation-based).

The next columns in Table 4 report the estimates by SMLE under different smoothing factors. Consistent with what we have seen in the Monte Carlo studies, estimates are sensitive to the smoothing factor λ . In particular, SMLE’s estimates for β and η are closer to NNE’s when λ is 3 or 7. However, SMLE’s estimates for δ are closer to NNE’s when λ is 10. This sensitivity to λ is problematic because, as discussed before, choosing the optimal λ is practically difficult.

Unlike in Monte Carlo studies, we do not know the true value of the search model parameter θ . Therefore, we cannot calculate the RMSE of either NNE or SMLE. So instead, we focus on model fit on three key statistics. Figure 8 gives the results. In each plot, the red dotted line shows the value observed in the real data. The histogram shows the predicted values by estimated search model. The histogram’s spread captures the statistical uncertainty in both $\hat{\theta}$ and the key statistic.

Figure 8 shows that NNE leads to a reasonable (albeit far from ideal) fit on the key statistics. With SMLE, the fit varies substantially by the smoothing factor λ . This sensitivity of SMLE to λ is problematic because, as discussed before, it is practically difficult to choose λ . More specifically, SMLE can give a good fit on either the buy rate or the number of searches, but not both. As to the search ranking, SMLE shows a substantially worse fit than NNE. Overall, NNE seems to give a better model fit.

The main findings from real data estimation are consistent with our findings in Monte Carlo studies. The root of the difficulty of SMLE lies in the enormous number of search and buy combinations. Without smoothing, an infeasibly large number of simulations will be needed to integrate out the unobservables for likelihood evaluation. Smoothing makes SMLE feasible, but it also introduces biases. In contrast, NNE does not require evaluating integrals over unobservables.